Java 登入畫面修正

https://gemini.google.com/app/9f68d6436ce84a45

User prompt: fix this code

Response: The original code has several major issues that prevent it from displaying or functioning properly:

1. **`setLayout(null)` without bounds**: Absolute layout is used without setting `setBounds(...)` for any component, making them invisible or default to size 0.
2. **String comparison with `==`**: In Java, string contents should be compared using `.equals()`, not `==`.
3. **Password input security**: `JTextField` is used for the password instead of `JPasswordField`.
4. **`setVisible(true)` order**: `setVisible(true)` is called before adding components, which can cause UI rendering artifacts.
5. **Missing close operation**: Closing the window won't terminate the Java process because `setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE)` is missing.

Here is the corrected and improved version using standard Swing layout managers (`GridLayout` / `BorderLayout`):

```
import javax.swing.*;
import java.awt.*;

public class GoodLogin extends JFrame {
    public GoodLogin() {
        setTitle("登入");
        setSize(320, 180);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null); // Center on screen

        // Main layout panel with padding
        JPanel mainPanel = new JPanel(new GridLayout(3, 2, 10, 10));
        mainPanel.setBorder(BorderFactory.createEmptyBorder(15, 15, 15, 15));

        JLabel l1 = new JLabel("帳號:");
        JTextField t1 = new JTextField();

        JLabel l2 = new JLabel("密碼:");
        JPasswordField t2 = new JPasswordField(); // Use password field for security

        JButton btn = new JButton("登入");

        mainPanel.add(l1);
        mainPanel.add(t1);
        mainPanel.add(l2);
        mainPanel.add(t2);
        mainPanel.add(new JLabel()); // Empty cell for grid alignment
        mainPanel.add(btn);

        add(mainPanel);

        btn.addActionListener(e -> {
            String username = t1.getText();
            String password = new String(t2.getPassword());

            // Correct String comparison using .equals()
            if ("admin".equals(username) && "123456".equals(password)) {
                JOptionPane.showMessageDialog(this, "登入成功！");
            } else {
                JOptionPane.showMessageDialog(this, "帳號或密碼錯誤！", "錯誤", JOptionPane.ERROR_MESSAGE);
            }
        });

        // Set visible AFTER components are added
        setVisible(true);
    }
```

```
    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> new GoodLogin());
    }
}
```